Evaluating Large Language Models for Clinical Decision-Making Using AgentClinic

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Introduction & Motivation

- **Limitations of Static Benchmarks:** Traditional medical datasets (USMLE, MedQA) assess factual recall using fully structured vignettes. They fail to evaluate dynamic hypothesis formation and adaptive questioning.
- **Clinical Reality:** Real-world diagnosis is a sequential, multi-turn, and uncertainty-driven process.
- **Objective:** Develop a highly modular multi-agent simulation framework (AgentClinic) to evaluate open-source LLMs under realistic, interactive clinical conditions.

Challenges in Multi-Agent Clinical Evaluation

- **Hardware Constraints:** Running multiple LLM agents (Doctor, Patient, Measurement) on local hardware poses severe VRAM bottlenecks.
- **Lexical Leakage:** In a homogeneous model setup, the Doctor and Patient share identical weights and latent embedding manifolds. This allows the Doctor to mathematically infer the Patient's ground-truth symptoms, artificially inflating accuracy.
- **Catastrophic Forgetting:** Medically fine-tuned datasets are narrow. Specialisation often degrades the model's broad conversational logic required to navigate interactive simulations.

AgentClinic v2.1: Enhanced Architecture

To address foundational flaws, the system was completely re-architected into AgentClinic v2.1:

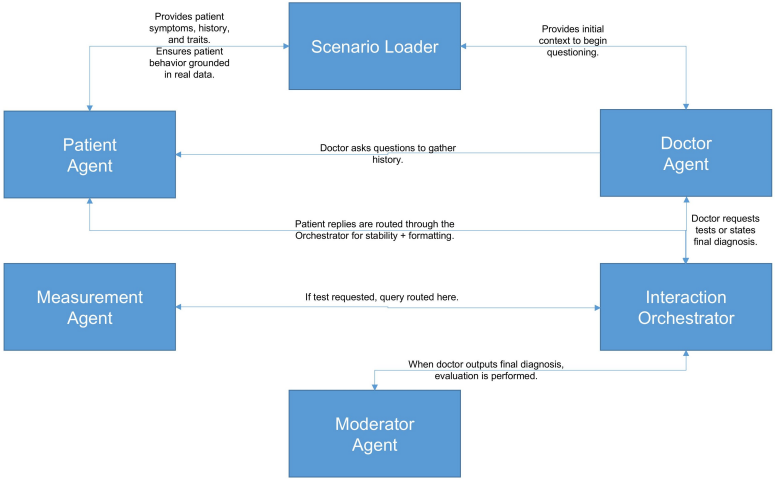
- **LangGraph DCG Orchestration:** Replaced the non-deterministic while-loop with a Directed Cyclic Graph for strict, state-machine-driven agent routing.
- **Heterogeneous Isolation:** Distinct, separate LLMs are used for Doctor vs. Patient to eradicate Lexical Leakage.
- **NF4 Double Quantisation:** Reduced 7B model footprint by 86%, safely enabling multiple heterogeneous engines efficiently on a single H100 GPU (95.8 GB VRAM).

The Metacognitive Reflection Node

Problem: Generating complex diagnostics and bedside dialogue simultaneously causes hallucinations (reasoning-contaminated speech).
Solution: The Latent Metacognitive Reflection (`doctor_reflection`) node.

- Forces the Doctor LLM to output a top-3 structured JSON differential diagnosis *before* every spoken utterance.
- Acts as a "think before you speak" latent Chain-of-Thought.
- Stored in the simulation trajectory for evaluation but kept hidden from the patient-facing dialogue history.

System Flowchart



Process-Oriented Clinical Metrics

To evaluate reasoning trajectory over terminal outcomes, three new metrics were formally defined via the `ClinicalTrajectoryEvaluator`:

- **Diagnostic Stability (DS):** Measures hypothesis thrashing using average Jaccard similarity across sequential dialogue turns.
- **Test Rationality (TR):** A compliance score marking whether non-invasive/basic exploratory tests were systematically prioritized over expensive/invasive ones.
- **Information Efficiency (IE):** The ratio of unique differential diagnoses explored per conversational turn.

Experimental Diagnostic Accuracy

Total Scenarios: 107 MedQA derived cases
Maximum Turns: 10

Configuration (AgentClinic v2.1)	Bias	Accuracy (%)
E1: Mistral-7B-v0.2 Baseline	None	45.6
E2: JSL-MedMistral-7B	None	54.2
E3: Llama-3.1-8B-Instruct	None	58.5

Key Finding: The reflection node logic entirely eclipsed the accuracy penalty of removing Lexical Leakage. Instruction-aligned reasoners (Llama-3.1) outperformed purely fine-tuned medical models.

The Safety Threat of Cognitive Biases

Ecologically valid distortions were injected to test model robustness. Under JSL-MedMistral-7B:

- **Recency Bias (Accuracy = 38.1%):**
  Destroys Diagnostic Stability ($DS = 0.45$). Instructing the model with recent $k = 5$ cases causes extreme hypothesis thrashing.
- **Confirmation Bias (Accuracy = 31.4%):**
  Artificially maximizes stability ($DS = 0.88$). Priming an initially incorrect hypothesis causes *premature diagnostic closure*, ignoring contradictory clinical evidence.

Conclusion: Deeply misaligned models can appear "confidently consistent" while being systematically incorrect.

Conclusion & Future Work

Conclusions:

- Clinical AI must be evaluated on its *process* (stability, rationality), not merely its terminal outcome.
- Structured metacognition scaffolds dramatically improve logical trajectory alignment.
- Open-source models remain highly vulnerable to systemic cognitive bias.

Future Directions:

- **Direct Preference Optimization (DPO):** Rewarding high DS/TR pathways.
- **Self-Play AI Feedback:** Using the trajectory evaluator as a programmatic reward signal for automated synthetic fine-tuning.